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Unit “ 4 : Session “ 6 : SLO “ 1 | Hardwired Control “ Choose the Best

■ I. Multiple Choice

1. b) Implemented with logic gates “ a hardwired control unit is pure digital logic (gates, decoders, flip-flops).
2. b) Decoder “ the instruction decoder interprets the opcode and drives signal generation.
3. b) Simple, fixed instruction sets “ hardwired logic is economical when the ISA is small and stable (RISC-style).
4. c) Faster operation “ signals come straight from logic gates with no control-memory read, so hardwired beats microprogrammed on speed.
5. c) Control memory “ control memory belongs to microprogrammed control, not hardwired.
6. b) Current instruction and processor state “ signals are functions of the opcode, step counter, and condition flags.
7. c) RISC processors “ their simple, fixed instruction sets make hardwired control practical; modern GPUs similarly use hardwired logic for speed.
8. b) Requires hardware modification “ the logic is physically wired, so adding an instruction means redesigning the circuit.
9. b) Hardwired control “ the control unit is designed as a finite state machine stepping through instruction states.
10. b) Propagation delay of logic gates “ the delay through the combinational logic sets the minimum cycle time.